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PROJECT AND TEAM INFORMATION

Project Title

Online Quiz Management System

Student/Team Information

Team Name: Team # (Mentor needs to assign)	Management
Team member 1 (Team Lead) Name :- Sachin Kumar Student Id:- 2520234694 Email:- sachin9973259472@gmail.com	
Team member 2 Name :- Rishu Kumar Student Id:- 2520230235 Email:- rishusatyam7970@gmail.com	
Team member 3 Name :- Sandeep Kumar Student Id:- 2520230181 Email:- sandeepkumar2003121100@gmail.com	

PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

The Quiz Management System is a web-based application designed to provide an interactive and efficient platform for conducting quizzes. The primary goal of this project is to simplify the process of quiz creation, participation, and evaluation for both administrators and users. It allows users to select a subject and difficulty level, attempt quizzes, and receive instant feedback based on their performance.

The system consists of two main modules: the admin module and the user module. The admin module enables administrators to add, update, and manage quiz questions, subjects, and difficulty levels. The user module allows participants to register or log in, choose their desired quiz category, and answer multiple-choice questions within a structured interface.

One of the key features of this system is automatic score calculation. The system evaluates user responses in real-time and displays the final score immediately after quiz submission. This reduces manual effort and ensures accuracy in evaluation. Additionally, the system maintains user data and performance records, which can be used for analysis and improvement.

The project is developed using modern web technologies, ensuring a user-friendly interface and smooth functionality. It is designed to be scalable and can be extended with additional features such as timer-based quizzes, leaderboards, and result analytics.

Overall, the Quiz Management System provides a reliable and efficient solution for online assessments, making it suitable for educational institutions and self-learning platforms.

Updated Project Approach and Architecture

The Quiz Management System follows a client-server architecture where the frontend interacts with the backend to process user requests and manage quiz data. The system is designed to be simple, modular, and easy to maintain.

On the frontend side, the application is developed using HTML, CSS, and JavaScript to create an interactive and user-friendly interface. Users can log in, select a subject and difficulty level, and attempt quizzes through dynamic web pages. JavaScript is used to handle user inputs, manage quiz flow, and store answers temporarily.

On the backend side, the system handles data processing, user authentication, and quiz evaluation. It communicates with the frontend using HTTP/HTTPS protocols. When a user submits answers, the frontend sends the data to the backend, where the system compares responses with correct answers and calculates the score.

The project uses a database to store user information, quiz questions, subjects, and results. Technologies like MySQL or similar databases are used for structured data storage. Queries are used to fetch and update data efficiently.

The system design follows a modular approach with separate components for user management, quiz management, and result processing. This improves scalability and makes it easier to add new features such as timers, leaderboards, and analytics.

Overall, the architecture ensures smooth communication between components, fast performance, and accurate result generation, making the system reliable for online quiz applications.

Tasks Completed

The Quiz Management System has successfully completed several important tasks that form the core functionality of the project. These completed tasks ensure that the system is working properly and ready for further enhancements.

Task Completed	Team Member
Project Setup and Basic Structure (HTML, CSS, JavaScript)	Sandeep Kumar
User Interface Design (Login, Quiz Pages, Layout)	Sandeep Kumar
Subject and Difficulty Selection Implementation	Sandeep Kumar
Quiz Flow Logic and Navigation	Sachin Kumar
Answer Storage using Data Structures (Arrays)	Sachin Kumar
Score Calculation and Result Logic	Sachin Kumar
Backend Processing and Data Handling	Sachin Kumar
Database Design and Integration	Rishu Kumar
Testing (Unit, Functional, UI Testing)	Rishu Kumar
Documentation Preparation	Rishu Kumar

Challenges/Roadblocks

During the development of the Quiz Management System, several challenges were faced at different stages of the project.

One of the major challenges was handling user responses and managing the quiz flow correctly. Initially, it was difficult to store and track answers for each question, especially when users navigate between questions. This issue was solved by using arrays in JavaScript to store answers dynamically and updating them as the user selects options. Another challenge was implementing accurate score calculation. Matching user answers with correct answers required proper indexing and validation. Errors occurred in the beginning due to mismatched indexes, but this was resolved by carefully aligning the question and answer arrays and testing with multiple cases.

Designing a user-friendly interface was also a challenge. Some input fields and layouts were not properly aligned, which affected usability. This was improved by adjusting CSS styles and making all input elements consistent in size and structure.

Database connectivity and data handling were also challenging. Ensuring that questions, users, and results are stored and retrieved correctly required proper query design. This was addressed by structuring the database tables clearly and testing queries step by step.

Another issue was maintaining smooth communication between frontend and backend. Sometimes data was not passed correctly during quiz submission. This was solved by debugging request-response handling and ensuring proper data formatting.

In the future, these challenges can be further handled by improving error handling, adding validations, and performing more testing. Overall, these problem-solving steps helped in making the system more stable and reliable.

Tasks Pending

Task Pending	Team Member (to complete the task)
Complete Admin Panel (edit/delete questions, full control dashboard)	<i>Sachin Kumar</i>
Implement Leaderboard / Ranking system	<i>Sachin Kumar</i>
Enhance UI/UX (responsive design, better layout)	<i>Sandeep Kumar</i>
Final Testing and Bug Fixing	<i>Rishu Kumar</i>

Project Outcome/Deliverables

The Quiz Management System successfully delivers a fully functional web-based application for conducting online quizzes. The main outcome of this project is an interactive platform where users can log in, select subjects and difficulty levels, attempt quizzes, and receive instant results based on their performance.

One of the key deliverables is the complete user module, which includes login functionality, quiz selection, question display, and answer submission. Another important deliverable is the admin module, which allows management of quiz questions, subjects, and difficulty levels.

The system also provides automatic score calculation and result display, ensuring quick and accurate evaluation. A structured database is implemented to store user data, questions, and results efficiently.

Additionally, the project includes a user-friendly interface, well-organized code, and proper system design, making it easy to understand and extend. Supporting documents such as project report, source code, and system architecture diagrams are also part of the final deliverables.

Overall, the project delivers a reliable and efficient solution for online quiz management, suitable for educational and learning purposes.

Progress Overview

The development of the Quiz Management System is currently in an advanced stage, with most of the core functionalities successfully implemented. The user module, including login, subject and level selection, quiz interface, and answer submission, has been completed. The score calculation feature is also working correctly, providing instant results after quiz completion.

The basic admin functionalities, such as adding and managing quiz questions, have been partially completed. However, some advanced features like full admin control panel, data editing options, and validation mechanisms are slightly behind schedule and require further improvement.

The frontend design and user interface are mostly completed and, in some aspects, ahead of schedule. Issues related to layout alignment and input field sizes have already been resolved, improving the overall user experience.

Database integration and basic data handling are functioning properly, but optimization and advanced query handling are still pending.

Overall, the project is progressing well, with major components completed. The remaining work mainly involves refining features, improving error handling, and adding optional enhancements like timers and leaderboards to make the system more robust and feature-rich.

Codebase Information

The project primarily uses the **main (or master) branch** for stable code. All major features and updates are committed to this branch after proper testing. Additional changes and experiments can be managed using separate branches if needed.

Several important commits have been made during development. Initial commits include project setup and basic UI structure using HTML, CSS, and JavaScript. Subsequent commits added core functionalities such as user login, quiz selection (subject and level), and question display. Further commits focused on implementing answer storage using arrays and developing the score calculation logic.

Recent commits include improvements in UI design, fixing input field sizes, and enhancing user interaction. Bug fixes related to incorrect answer mapping and data handling were also addressed in later commits.

Overall, the codebase is organized, regularly updated, and maintained with meaningful commit messages, making it easy to track progress and manage future enhancements.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Unit Testing	Pass	Individual functions like score calculation and answer storage tested successfully
Integration Testing	Pass	Frontend and backend communication working correctly
UI Testing	Pass	Layout issues fixed, input fields aligned properly
Functional Testing	Pass	Login, quiz selection, and submission working as expected
Database Testing	Pass	Data storage and retrieval functioning correctly
Performance Testing	Partial Pass	Works well for small data, needs optimization for large data
Error Handling Testing	Partial Pass	Basic validations added, but needs improvement for edge cases

Deliverables Progress

The development of the Quiz Management System is progressing steadily, with most of the key deliverables either completed or in the final stages.

The **user module** is completed, including login functionality, subject and level selection, quiz interface, and answer submission. The **score calculation and result display** feature is also completed and working correctly.

The **admin module** is currently in progress. Basic functionality for adding questions has been implemented, but features like editing, deleting, and full control panel are still under development.

The **database design and integration** are mostly completed. Data for users, questions, and results can be stored and retrieved successfully, though some optimization is still pending.

The **user interface (UI design)** is completed, with improvements made in layout and input field consistency.

The **system architecture and documentation** are in progress. Basic structure and diagrams are prepared, but final documentation needs refinement.

Additional features such as **timer, leaderboard, and advanced validation** are currently pending and planned for future enhancement.

Overall, the majority of core deliverables are completed, while a few advanced features and documentation tasks are still in progress or pending.